

Rational translated-Wick normal form and payload-comparable reduction

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

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1 Purpose and scope

R-085 reduced the nonlinear rational Pauli–Fierz row to the signed packet

$$\mathcal{H}_R + \frac{1}{2}c^T \overline{B}(U + a)c, \quad (1.1)$$

where $\mathcal{H}_R$ is the shifted-Hessian pair. This note proves three further facts about (1.1).

First, the three displayed terms in R-085 are one exact translated-Wick normal form. If L is the second-order Taylor remainder of the heat-lifted Gram, then

$$\boxed{\mathcal{H}_R + \frac{1}{2}c^T \overline{B}(U + a)c = \frac{1}{2}L : \{(G + c)^{\otimes 2} - \Gamma\} + \frac{1}{2}c^T B_T c,} \quad (1.2)$$

where B_T is the quadratic Taylor polynomial at U . This is an identity, not a positivity statement.

Second, all coefficient-nonresonant and payload-comparable dyadic pieces of (1.1) are form-payable by the accepted R-063, R-068, R-071, R-076, and R-078 estimates. The Q branch retains the R-085 quadratic-payload total 7/10; the G branch retains the R-078 $a^2 Da$ total 23/30 and moment 30/7. An independent base-frozen cubic coordinate has total 13/15, slack 2/15, and moment 15/2. Thus the whole rational question is reduced to one coefficient-dominant high–high-to-low shifted resonance, with the endpoint square and Wick trace retained.

Third, exact production fixtures show why the remaining coupling cannot be split. The Taylor Gram B_T can be negative. A two-coordinate endpoint kernel makes the rational square vanish while the cross remainder is affine and unbounded below in a free derivative jet. Small heat convolution lifts that kernel only by $O(\sigma^2)$, while the cross remainder stays nonzero, so an inverse Schur estimate loses $O(\sigma^{-2})$. These are method obstructions only: Cameron–Martin derivative energy and the complete backward-heat/trace packet are absent from the local fixtures.

No complete rational NEAR bound, production Cartan atom estimate, controlled Cartan CFAR, REG, OVERLAP, progressive theorem, controlled-shell one-use estimate, Nelson moment, measure theorem, or Sector-A closure is asserted. The tier remains T4.

2 Authorities and exact variables

The load-bearing authorities are R-050 for the production current, R-063 for the balanced coefficient-jet models, R-068 for centered-form duality, R-071 for smooth matrix one-forms, R-076–R-078 for input-maximum payloads and the coefficient-dominant taxonomy, and R-084–R-085 for the heat-lifted endpoint and five-family payment.

Write

$$B_0 = \overline{B}_\Sigma(U), \quad B_1 = \overline{B}_\Sigma(U + a), \quad G = DU, \quad c = Da, \quad (2.1)$$

$$Q = G^{\otimes 2} - \Gamma, \quad Q^a = (G + c)^{\otimes 2} - \Gamma. \quad (2.2)$$

The production rational map and Gram are

$$g(z) = P_{\text{dbl}} z - \frac{5}{9} \frac{|P_{\text{dbl}} z|^2}{|z|^2 + \epsilon_\rho} z, \quad B_R(z) = 4c_1 g(z) \otimes g(z), \quad (2.3)$$

$$c_1 = \frac{243}{8000(4 + 10^{-12})}, \quad \overline{B}_\Sigma = P_\Sigma B_R. \quad (2.4)$$

All algebra below is finite-cutoff and pathwise. The form estimates are on the regular mutually orthogonal strict-past one-shot class used in R-080–R-085. The density floor is fixed and positive.

3 Exact translated-Wick normal form

Put

$$K_t = D^2 \overline{B}_\Sigma(U + ta) - D^2 \overline{B}_\Sigma(U) \quad (3.1)$$

and define

$$L = \int_0^1 (1-t) K_t[a, a] dt = B_1 - B_0 - D\overline{B}(U)[a] - \frac{1}{2} D^2 \overline{B}(U)[a, a]. \quad (3.2)$$

The equality is Taylor's theorem in the opposite integration order. The same remainder also has the cubic coordinate

$$L = \frac{1}{2} \int_0^1 (1-t)^2 D^3 \overline{B}(U + ta)[a, a, a] dt. \quad (3.3)$$

Let

$$B_T = B_0 + D\overline{B}(U)[a] + \frac{1}{2} D^2 \overline{B}(U)[a, a], \quad (3.4)$$

so $B_1 = B_T + L$. R-085's shifted-Hessian pair is exactly

$$\mathcal{H}_R = \frac{1}{2} L : Q + G^T L c. \quad (3.5)$$

Since

$$\frac{1}{2} L : Q^a = \frac{1}{2} L : Q + G^T L c + \frac{1}{2} c^T L c, \quad (3.6)$$

adding $c^T B_1 c/2$ and using $B_1 - L = B_T$ proves (1.2):

$$\mathcal{H}_R + \frac{1}{2} c^T B_1 c = \frac{1}{2} L : Q^a + \frac{1}{2} c^T B_T c. \quad (3.7)$$

The importance of (3.7) is negative as well as positive. It identifies the right stochastic object as an adapted translated-Wick third remainder, but it does not license separate lower bounds for its two terms. Section 7 gives exact cancellation fixtures.

4 A sharp cubic Sobolev payment

Let a be a finite-dimensional vector field on the fixed three-torus, with $a \in H^2 \cap L^6$. Tensor contractions and finite component sums are suppressed.

Theorem 4.1

For $0 \leq s \leq 2$,

$$\|a^{\otimes 3}\|_{H^s} \leq C_s \|a\|_{H^2}^s \|a\|_6^{3-s}. \quad (4.1)$$

Proof. At the low endpoint,

$$\|a^{\otimes 3}\|_2 \leq C \|a\|_6^3. \quad (4.2)$$

At the high endpoint, the three-dimensional Gagliardo–Nirenberg inequality gives

$$\|a\|_\infty^2 \leq C \|a\|_{H^2} \|a\|_6. \quad (4.3)$$

The zeroth and first derivatives are bounded by the right side below using $H^2 \hookrightarrow W^{1,6} \cap L^6$. For the second derivative, the only types are $a^2 D^2 a$ and $a(Da)^2$, and hence

$$\begin{aligned} \|a^{\otimes 3}\|_{H^2} &\leq C \{\|a\|_\infty^2 \|a\|_{H^2} + \|a\|_6 \|Da\|_6^2\} \\ &\leq C \|a\|_{H^2}^2 \|a\|_6. \end{aligned} \quad (4.4)$$

Sobolev interpolation of the fixed tensor $a^{\otimes 3}$ between (4.2) and (4.4) proves (4.1). $\square$
Put

$$X = \|a\|_{H^2}^2, \quad Y = \|a\|_6^6. \quad (4.5)$$

If M_s is a control-independent H^{-s} model norm, (4.1) yields

$$|\langle \mathbb{Q}, a^{\otimes 3} \rangle| \leq C M_s X^{s/2} Y^{(3-s)/6}. \quad (4.6)$$

At $s = 11/10$ the exponent ledger is

$$\frac{s}{2} = \frac{11}{20}, \quad \frac{3-s}{6} = \frac{19}{60}, \quad \frac{11}{20} + \frac{19}{60} = \frac{13}{15}, \quad (4.7)$$

with slack $2/15$. One three-factor weighted Young inequality therefore gives

$$M_{11/10} X^{11/20} Y^{19/60} \leq \eta X + \zeta Y + C \eta^{-33/8} \zeta^{-19/8} M_{11/10}^{15/2}. \quad (4.8)$$

The R-063 balanced coefficient-jet theorem supplies every finite moment of the base-frozen decorated model $D^3 \overline{B}(U) \diamond Q$. Thus the base-frozen part of (3.3) closes with moment $15/2$. Formula (4.8) is sharper than multiplying the old single-control ledger three times, which would give the false impression of the supercritical total $21/20$.

The qualification “base-frozen” is load-bearing. R-063’s translation is for deterministic H^2 shifts. It does not turn $D^3 \overline{B}(U + ta) - D^3 \overline{B}(U)$ under an adapted random shift into a control-independent model norm.

5 Dyadic closure of the payable orientations

Use one smooth inhomogeneous analysis partition $(\Delta_m)_{m \geq -1}$. Assign every monomial in $a \otimes a$ and $a \otimes a \otimes c$ to its largest control input, with a fixed tie rule:

$$a \otimes a = \sum_r P_r, \quad a \otimes a \otimes c = \sum_r H_r. \quad (5.1)$$

Every P_r and H_r is supported in a ball of radius $C2^r$, including high-high-to-low output. Expand the finite-cutoff contractions in (3.5). The only dangerous Bony orientation is a resonance between the shifted coefficient and the Gaussian model. With a fixed resonance width L_0 , split it at the payload scale:

$$T_Q^{\leq} = \frac{1}{2} \int_0^1 (1-t) \sum_r \sum_{\substack{|m-n| \leq L_0 \\ m \leq r+L_0}} \int (\Delta_m K_t)[P_r] : \Delta_n Q \, dx \, dt, \quad (5.2)$$

$$T_G^{\leq} = \int_0^1 (1-t) \sum_r \sum_{\substack{|m-n| \leq L_0 \\ m \leq r+L_0}} \int \langle \Delta_n G, (\Delta_m K_t)[H_r] \rangle \, dx \, dt. \quad (5.3)$$

Equality belongs to the $\leq$ branch. The two nonresonant paraproducts are included with (5.2)–(5.3) in the payable packet.

At fixed floor,

$$\sup_{t,z} \|D^2 \bar{B}(z+ta) - D^2 \bar{B}(z)\| \leq 2 \sup_z \|D^2 \bar{B}(z)\| < \infty. \quad (5.4)$$

For the Q branch, the direct block sum uses the R-068 $H^{-1-\kappa}$ centered model and the quadratic input-maximum theorem. At $\kappa = 1/10$ it has

$$\boxed{|T_Q^{\leq}| \lesssim M_Q X^{11/20} Y^{3/20}}, \quad \frac{11}{20} + \frac{3}{20} = \frac{7}{10}. \quad (5.5)$$

For the G branch, R-078's exact quadratic-payload Besov theorem gives at $s_G = 3/5$

$$\boxed{|T_G^{\leq}| \lesssim M_G X^{2/5} Y^{11/30}}, \quad \frac{2}{5} + \frac{11}{30} = \frac{23}{30}. \quad (5.6)$$

The corresponding slacks are $3/10$ and $7/30$; the worse required moment is $30/7$. All these moments are accepted. Summing the shell blocks before Young spends one control allocation, one sextic allocation, and one random constant. No adapted multiplier norm is asserted.

Theorem 5.1

For every $\eta, \zeta > 0$, the sum $\mathcal{P}_R^{\text{pay}}$ of the two nonresonant orientations, (5.2), (5.3), and the base-frozen cubic model obeys

$$\boxed{\mathbb{E} \mathcal{P}_R^{\text{pay}} \geq -\eta \mathbb{E} X - \zeta \mathbb{E} Y - C_{\eta, \zeta, \epsilon_\rho, L_0}}. \quad (5.7)$$

The constant is uniform in the terminal cutoff on the declared regular one-shot class. The proof is (4.8), (5.5), (5.6), followed by one global weighted Young allocation. The finite number of Bony orientations changes only the constant.

6 The single surviving coefficient-dominant packet

The exact complements of (5.2)–(5.3) are

$$T_Q^> = \frac{1}{2} \int_0^1 (1-t) \sum_r \sum_{\substack{|m-n| \leq L_0 \\ m > r+L_0}} \int (\Delta_m K_t)[P_r] : \Delta_n Q \, dx \, dt, \quad (6.1)$$

$$T_G^> = \int_0^1 (1-t) \sum_r \sum_{\substack{|m-n| \leq L_0 \\ m > r+L_0}} \int \langle \Delta_n G, (\Delta_m K_t)[H_r] \rangle \, dx \, dt. \quad (6.2)$$

They are high–high-to-low forms: the shifted coefficient and Q or G are comparable and strictly higher than every explicit payload input. No frequency separation follows from their product, since opposite carriers can create the zero mode.

Combining Theorem 5.1 with the exact R-085 endpoint leaves precisely

$$\boxed{\mathcal{P}_R^> = T_Q^> + T_G^> + \frac{1}{2} c^T \bar{B}(U + a)c.} \quad (6.3)$$

The rational NEAR successor is a lower form bound for (6.3), with the Wick trace and lower-chaos forest retained. There is no remaining list of five unclassified rational families.

A first parilinearisation separates a base U principal, an a principal, and commutators. The sharper cubic theorem pays the base-frozen coordinate, but extracting the fully shifted a -dependent coefficient as a random multiplier is forbidden: its high–high-to-low resonance is exactly (6.1)–(6.2). The next proof must use a largest-root/Doob or backward-heat cancellation on the complete packet rather than another absolute coefficient estimate.

7 Exact inseparability diagnostics

The scalar Taylor Gram is indefinite

Remove the harmless positive factor $4c_1$ and put

$$f_e(x) = x - \frac{5}{9} \frac{x^3}{x^2 + e}, \quad B(x) = f_e(x)^2. \quad (7.1)$$

At $z = \sqrt{e}$ and $a = C\sqrt{e}$, direct differentiation gives

$$\boxed{B_T = \frac{e}{324} (169 + 208C - C^2).} \quad (7.2)$$

Thus $B_T < 0$ when

$$C < 104 - 13\sqrt{65} \quad \text{or} \quad C > 104 + 13\sqrt{65}. \quad (7.3)$$

At either $C = -1$ or $C = 209$,

$$B_T = -\frac{10e}{81}, \quad B_{R,T} = -\frac{40c_1 e}{81}. \quad (7.4)$$

For $C = -1$ the endpoint is zero, so $B_1 = 0$ and $L = -B_T$. In a local jet with $G = 0$, $\Gamma = 0$, the two terms of (3.7) cancel exactly for every c . Treating either one in isolation creates a false sign.

Endpoint-kernel leakage

On the doublet/singlet slice take

$$z = (1, 1), \quad a = (1, -1), \quad z + a = (2, 0). \quad (7.5)$$

At $e = 0$, for the normalized Gram $g \otimes g$,

$$B_1 = \frac{1}{81} \begin{pmatrix} 64 & 0 \\ 0 & 0 \end{pmatrix}, \quad B_T = \frac{1}{81} \begin{pmatrix} -1 & 5 \\ 5 & 0 \end{pmatrix}, \quad (7.6)$$

$$L = \frac{1}{81} \begin{pmatrix} 65 & -5 \\ -5 & 0 \end{pmatrix}. \quad (7.7)$$

At positive floor,

$$L_{21} = \frac{5(27e^2 + 40e - 8)}{81(e + 2)^3} < 0 \quad (7.8)$$

for $e = 10^{-12}$. Take $G = e_1$, $c = te_2$, and $\Gamma = 0$. Because $e_2 \in \ker B_1$, the endpoint c square and its cross vanish, but the packet contains tL_{12} and is unbounded below as $t \rightarrow +\infty$. This refutes pointwise and derivative-cost-free rational Schur absorption. It is not a spatial form counterexample because a field realizing large Da pays Cameron–Martin energy.

Heat ellipticity degenerates at the endpoint

Let $P_{\sigma^2 I}$ be isotropic value heat convolution. At $(2, 0)$ the kernel vector is e_2 , and

$$\partial_y g_2(2, 0) = -\frac{20}{9(4 + e)}. \quad (7.9)$$

Consequently

$$e_2^T (P_{\sigma^2 I} B_R)(2, 0) e_2 = 4c_1 \sigma^2 \left\{ \frac{20}{9(4 + e)} \right\}^2 + O(\sigma^4). \quad (7.10)$$

The cross coefficient in (7.8) approaches a nonzero constant while this eigenvalue is $O(\sigma^2)$. A direct inverse-Gram Schur complement therefore costs $O(\sigma^{-2})$. Since $\Sigma = 0$ is an actual heat endpoint, heat positivity alone is not uniform. Any successful heat argument must cancel this loss with the backward-heat/trace packet.

The three diagnostics fire NG-2026-07-25-A13-RATIONAL-TRANSLATED-WICK-SEPARATION-AND-HEAT-SCHUR. They forbid Taylor-Gram positivity, endpoint-square-only absorption, and uniform inverse-heat-Gram Schur closure. They do not refute (6.3).

8 Updated completion map

Rational NEAR. The five R-085 families and the R-086 payable shifted orientations are closed on the regular one-shot class. Prove (6.3) by a complete coefficient-dominant largest-root/Doob or backward-heat argument.

Cartan CFAR. The R-085 production atom estimates (4.10)–(4.11), with exponent $s > 1/2$, remain open. Nothing in the rational reduction supplies spatial Cartan off-diagonal decay.

REG. After rational NEAR and Cartan CFAR, reassemble R-080 low objects, R-083 polynomial CFAR, R-084 linear rows, traces, forest terms, and paid subtraction into the complete regular packet lower bound.

OVERLAP and CORE. REG does not imply a partition/revisit-uniform bound. OVERLAP remains a separate complete-packet theorem. A graph-core extension may be proved only after that lower bound exists; it cannot repair a missing analytic estimate.

One-use and Nelson. Only after the all-progressive lower bound may the 0.45 control allocation and 0.15 sextic allocation be inserted into Boue–Dupuis. The target $q = 10/9$ remains conditional.

9 Proof-and-failure map

route	verdict and target
five unshifted families	closed in R-085
translated-Wick form	exact, (3.7)
base-frozen cubic Q	closed, 13/15, $M^{15/2}$
nonresonant shifted	closed, Theorem 5.1
payload-comparable	closed, 7/10, 23/30
Taylor Gram positive	failed, $B_T = -10e/81$
endpoint-square Schur	failed, $\ker B_1$ leakage
uniform heat Schur	failed, $\lambda_{\min} = O(\sigma^2)$
coefficient-dominant	open, $\mathcal{P}_R^>$
rational NEAR	open, requires $\mathcal{P}_R^>$
Cartan atom	open, R-085 (4.10)–(4.11)

The map is cumulative. Future work starts at (6.3), not at the already paid five families or the three refuted pointwise Schur routes.

10 Devil’s-advocate review

1. **“The exact normal form proves positivity.” UPHELD as false.** Equation (7.2) makes B_T negative. Equation (3.7) is useful because it enforces coupling, not because either term has a sign.
2. **“The cubic Sobolev theorem closes every shifted multiplier.” UPHELD as false.** It closes a control-independent decorated model and the declared payable dyadic orientations. R-063 does not authorize adapted random translation of its model norm; (6.1)–(6.2) remain.
3. **“A bounded Hessian difference is a harmless multiplier.” UPHELD as false.** Boundedness closes only coefficient-nonresonant and payload-comparable blocks. A bounded high coefficient resonating with a rough model can create a zero mode.
4. **“Heat convolution makes the Gram uniformly invertible.” UPHELD as false.** Equation (7.10) gives only $O(\sigma^2)$ lifting at the actual zero-heat endpoint, so the inverse costs $O(\sigma^{-2})$.
5. **“Kernel leakage disproves the desired spatial form bound.” DISMISSED.** The local derivative jet omits Cameron–Martin H^2 energy and the complete Wick/heat/forest packet. It refutes a method, not the theorem.
6. **“Rational progress also closes Cartan CFAR.” UPHELD as false.** The two gates use different stochastic and spatial mechanisms. The complete mixed Cartan atom still needs decay $s > 1/2$.

7. “REG can now be promoted.” UPHELD as false. Both (6.3) and the Cartan atom remain open, so complete regular reassembly has not started.

11 Executable verification

The primary executable derives the rational parameters from the A1 manifest, checks the exact normal form on symbolic matrices, derives the cubic Sobolev and Young ledgers, proves the scalar Taylor formula, evaluates the exact two-coordinate kernel matrices and positive-floor sign, and derives the heat kernel coefficient. It leaves every downstream gate false.

The non-importing audit uses high-precision directional differentiation, independent finite differences for the normal form, numerical Gaussian heat quadrature at several variances, and separately recomputed exponent ledgers. It imports no code from the primary executable.

Run from the repository root:

```
python codes/foundations/  
a13_classii_rational_translated_wick_payload_comparable_  
reduction_verify.py
```

The integrated verifier pins the authority and evidence hashes, reruns both children, compares their common quantities, checks this PDF and every status surface, and fails closed if the coefficient-dominant packet, complete rational NEAR, Cartan CFAR, REG, OVERLAP, CORE, one-use, Nelson, or Sector A is promoted.

Result footer

Result ID: A13-CLASSII-RATIONAL-TRANSLATED-WICK-PAYLOAD-
COMPARABLE-REDUCTION

Precise statement:	The R-085 shifted-Hessian pair plus endpoint square has one exact translated-Wick normal form. A sharp cubic Sobolev theorem pays its base-frozen Q model with total 13/15 and moment 15/2. All nonresonant and payload-comparable shifted branches are form-paid; their Q/G totals are 7/10 and 23/30. The sole rational remainder is a coefficient-dominant high-high-to-low packet coupled to the endpoint square.
Scope:	Fixed positive floor; finite cutoff; common real-even regulator; exact products; regular mutually orthogonal strict-past one-shot controls.
Dependencies:	A1; R-050; R-063; R-068; R-071; R-076--R-085.
Evidence grade:	ANALYTIC, EXACT, EXECUTED.
Reproduction command:	python codes/foundations/a13_classii_rational_translated_wick_payload_comparable_reduction_verify.py
Expected output:	Primary, independent, integrated, and aggregate manifest-pinned PASS; exit zero.
Negative result:	NG-2026-07-25-A13-RATIONAL-TRANSLATED-WICK-SEPARATION-AND-HEAT-SCHUR (method no-go only).
Falsification gate:	Failure of any normal-form identity, exponent, fixture, heat asymptotic, hash, PDF, surface, or assertion contract.
Tier before / after:	T4 / T4; no promotion.
No-overclaim statement:	No coefficient-dominant rational packet bound, complete rational or signed NEAR, production Cartan atom, controlled Cartan CFAR, REG, OVERLAP, CORE, one-use, Nelson bound, measure theorem, removal limit, or Sector-A closure.
Next required action:	Prove the coefficient-dominant packet (6.3) by a largest-root/Doob or backward-heat/trace estimate; independently prove R-085 (4.10)--(4.11), then REG.